

CHAPTER 16

DERIVATIVES OF HYDROCARBONS

It was once believed that we could obtain carbon containing substances such as sugar, starch, camphor, urea etc. only from living organisms because only these organisms contained the 'vital force' necessary for their synthesis. Therefore, the branch of chemistry which deals with these compounds was known as organic chemistry. But in 1828 German chemist Friedrich Wohler used ammonium cyanate, an inorganic compound, to synthesize urea, an organic compound found in human urine. The laboratory synthesis of urea proved that vital force was not needed in the production of organic compounds. This was the first time that an organic compound (urea) was synthesized from an inorganic compound (ammonium cyanate.)



Friedrich Wohler

Friedrich Wohler (1800–1882), a German chemist, is known for being the first person to synthesize an organic compound from an inorganic compound. By making urea in the laboratory, he dealt a mortal blow to the vitalistic force theory which said that a vital force was necessary to produce the compounds found in living organisms.
Along with Liebig, Wohler studied the oil of bitter almonds and worked with other scientists on finding a method for the separation of silicon and beryllium from their mixtures. He also played an important role in the synthesis of calcium carbide and silicon nitride.

Carbon chemistry is the branch of chemistry which deals with the study of carbon and its compounds. We know that carbon atoms are capable of forming long chains by bonding to each other, a property known as catenation. Catenation is one of the reasons why such a large number of carbon compounds are found in nature. Of the compounds known today, almost 96% contain carbon.

We know that the compounds that contain only hydrogen and carbon are known as hydrocarbons. If all four valences of carbon in any compound are satisfied through single bonds, the compound is called alkane and is a saturated hydrocarbon. If they are satisfied through double or triple bonds then compounds are alkenes and alkynes respectively and are also known as unsaturated hydrocarbons.

In class 9th, we studied alkanes in some detail. Alkanes are depicted by the general formula RH. When a hydrogen atom is removed from the alkane then the group (radical) formed is referred to as alkyl group and is generally depicted by –R. For example, when a hydrogen atom is removed from methane (CH_4) we get methyl ($–CH_3$) and it is removed from ethane we get ethyl ($–C_2H_5$) group. Are the properties of carbon compounds determined by these alkyl group or by some other group? Let us explore.

A functional group is an atom or group of atoms which when joined to an alkyl group determines the characteristic chemical properties of that carbonic compound. Some functional groups are shown in table–1.

Table–1: Some functional groups seen in carbonic compounds

Name of functional group	Formula
Aldehyde	O ‖ –C–H
Ketone	>C=O
Carboxylic acid	O ‖ –C–OH
Halo (chloro, bromo etc.)	–Cl, –Br

Questions

1. Identify and write the names of the functional groups in the following compounds

   C_3H_7OH, C_4H_9Cl, CH_3CHO, $C_5H_{11}COOH$

2. Write the names of the compounds formed by joining –OH functional group to ethyl and propyl groups respectively.
3. Name the compounds formed when –Br and –COOH are added to methyl groups.

You must have heard about caustic soda. The chemical formula of caustic soda is NaOH which means that sodium is joined to –OH group. Let us see another compound where –OH is present.

If one hydrogen atom of methane (CH_4) is replaced by –OH group then we get methyl alcohol. Now we know two compounds where the –OH group is present. These are:

NaOH and CH_3OH

Draw the electron dot structures of the two compounds given above. What is the type of bond joining Na^+ and –OH in NaOH and that between $–CH_3$ and –OH?

We looked at the structural differences between NaOH and CH_3OH. Let us now understand the nature of the two compounds through an activity.

Activity–1

- Take two beakers or glasses.
- Take 5 mL water in the first beaker and add a sodium hydroxide pellet to it.
- Take 5 mL methyl alcohol in the second beaker.
- Test the solutions in the two beakers with litmus paper and universal indicator and fill in your observations in the table given below.

Table–2: Nature of sodium hydroxide and methyl alcohol

Substance	Effect on red litmus paper	Effect on blue litmus Paper	pH according to universal indicator
Solution of sodium hydroxide			
Methyl alcohol			

From our observations in Activity–1 and our understanding of the nature of chemical bond (ionic bond in sodium hydroxide and covalent bond in methyl alcohol), we can see that both the compounds have an –OH group but their natures are different. NaOH has hydroxide ion (OH^- ion) and CH_3OH has hydroxyl group (–OH). Since methyl alcohol is a carbonic compound therefore –OH acts as a functional group. Therefore, the nature of the two substances also depends on the root group to which OH is attached.

16.1 Alcohol

When an alkyl group bonds with –OH (a functional group) then the resulting compound is known as alcohol or alkyl alcohol or alkanol. Alkanols are also known as hydroxyl derivative of hydrocarbons.

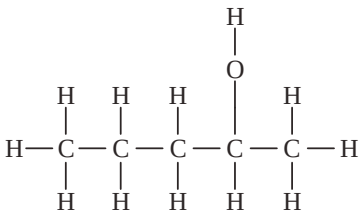
16.1.1 Nomenclature of alcohols

According to IUPAC nomenclature, the suffix –e in the parent hydrocarbon is replaced by the suffix –ol when naming alcohols. Table–3 has some examples; use them to replace the question marks with the correct alcohol names.

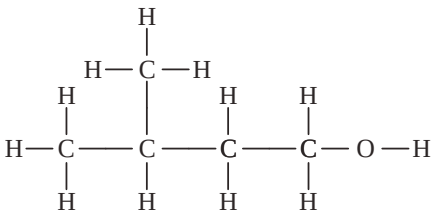
Table-3

Parent hydrocarbon	IUPAC name	Formula
Methan***e***	Methan***ol***	CH_3OH
Ethan***e***	Ethan***ol***	CH_3CH_2OH
Propane	?	?
?	Butanol	?

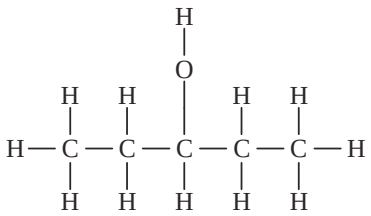
Are the four structures shown below same or different? Name the compounds.



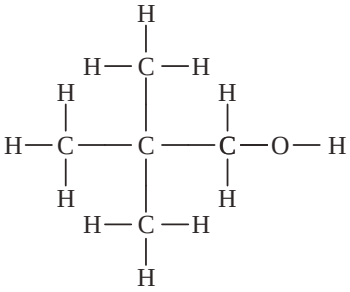
(a)



(b)



(c)

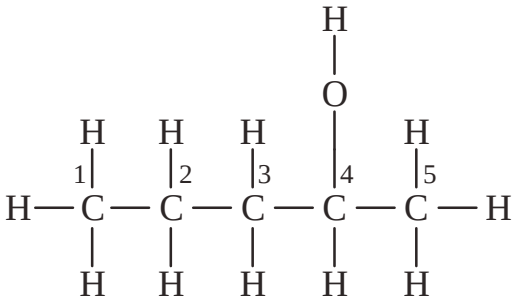


(d)

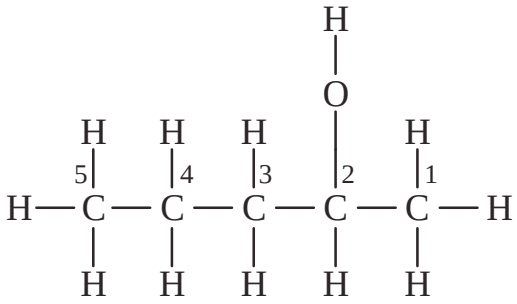
We find that all four compounds have the same molecular formula, $C_5H_{12}O$ but their structural formulae are different. We learnt the IUPAC nomenclature of hydrocarbons in class-9th. For naming alcohols according to IUPAC:

- Choose the longest, continuous carbon chain having –OH (alcohol) functional group.
- The longest chain of carbon atoms containing the –OH group is numbered in such a way that the –OH group is attached at the carbon atom possessing lowest possible number in the chain.
- Now follow the steps you used while naming hydrocarbons. Write the position number of alkyl group followed by the name of the parent chain followed by a dash, then the position number of the –OH group followed by a dash and the suffix –ol.

Let us now use these rules to name the structures shown above. In structure 'A', the carbon chain can be numbered in two ways as shown below:



A (i)



A (ii)

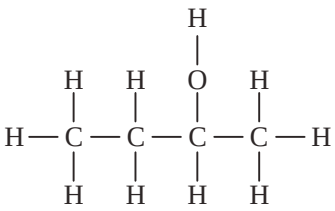
Which numbering is correct, A (i) or A (ii)? According to IUPAC rules, the numbering shown in A(ii) is correct because in A (ii) the carbon to which the –OH group is attached is numbered 2 where as in A (i) it is numbered 4. Therefore, the structure shown in A(ii) is pentan–2–ol.

If we examine 'A' and 'C' we find that their molecular formula are same but the locations of the carbon atoms to which –OH groups are attached are different. Compounds whose molecular formula is same but which differ from each other in the location of the functional groups on or in the carbon chain are called positional isomers.

Similarly, if look carefully at 'B' and 'D', we find that the carbon chains of the two molecules are different although their molecular formula is same. Compounds whose molecular formula is same but which differ from each other in the arrangement of the carbon chain are called chain isomers.

Questions

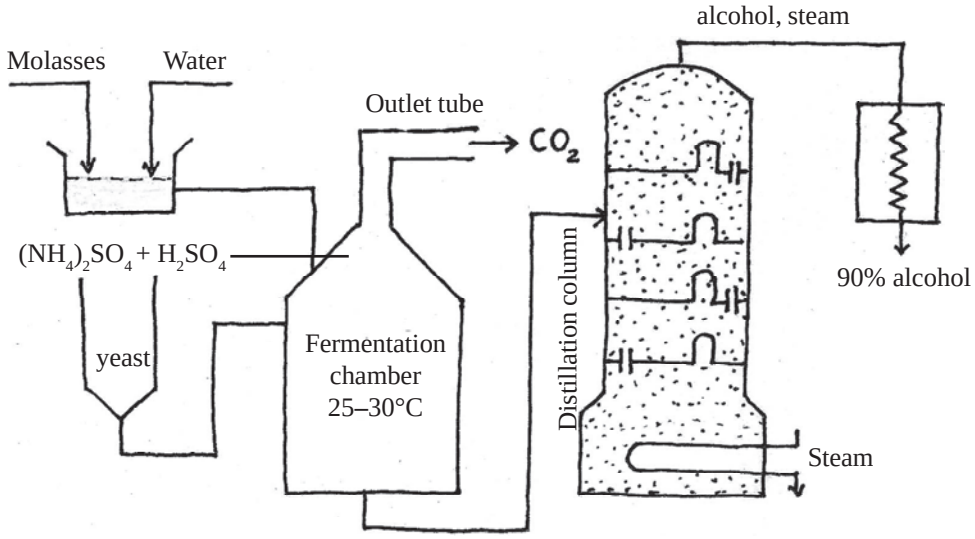
1. Write the names of two compounds having hydroxyl group and two having hydroxide group.
2. Write the IUPAC name of the given compound.



3. Write the structural formula of 2–methylpropan–1–ol.
4. Write the structural formulae of all possible isomers of C_3H_8O.

16.1.2 Industrial manufacture of ethanol

Industrial manufacture of ethanol (ethyl alcohol) is done using fermentation method. In this process, sugar is first separated from cane–sugar juice by crystallization. The thick, yellow syrupy substance left behind is called molasses. Molasses are diluted by adding water to obtain a 8–10% solution. Then ammonium sulphate, which promotes yeast reproduction, is added to the solution. A small amount of sulphuric acid is also added to the solution. Sulphuric acid also promotes the growth of yeast and kills other microorganisms.



Figure–1: Industrial manufacture of ethanol from molasses

The mixture is filled in a tank (as shown in figure1) and 5% yeast is added to the tank and the mixture is left for two-three days, in the absence of air, at 25-30°C. Yeast is a type of fungus which has the enzymes invertase and zymase. Invertase converts the sucrose present in molasses into glucose and fructose. Ethanol is obtained by fermentation of glucose and fructose in the presence of zymase. In this process, carbon dioxide gas is released with rapid bubbling and is removed via the outlet tube. We can show the formation of alcohol by the following equation:

$$\underset{\textit{Sucrose}}{C_{12}H_{22}O_{11} + H_2O} \xrightarrow{\text{Invertase}} \underset{\textit{Glucose}}{C_6H_{12}O_6} + \underset{\textit{Fructose}}{C_6H_{12}O_6}$$

$$C_6H_{12}O_6 \xrightarrow{\text{Zymase}} \underset{\textit{Ethanol}}{2C_2H_5OH} + 2CO_2\uparrow$$

Biocatalysts : Enzymes – Catalysts are substances that do not themselves take part in a chemical reaction but whose presence affects the rate of the reaction. Catalysts that effect biological processes and reactions are called enzymes or biocatalysts.

The alcohol obtained by this method is quite dilute. 100% alcohol, that does not have any trace of water, is called absolute alcohol. Ethanol which has 5% water is called rectified spirits. The mixture of petrol and ethanol is called power alcohol and it is used as a fuel in automobiles.

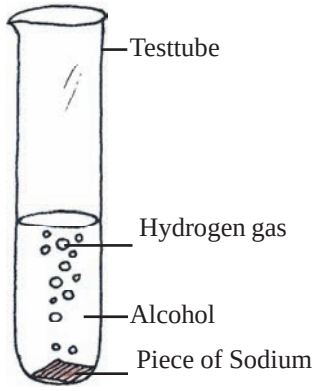
16.1.3 Properties of alcohols

Usually alcohols range from weakly acidic to neutral in nature. Let us understand this through an activity.

Activity–2

- Take 10 mL ethanol in a testtube.
- Cut a rice grain–sized piece of sodium from kerosene and dry it on a filter paper.
- Add this sodium to the ethanol in the testtube and observe and record what happens.

Ethanol reacts with extremely reactive metals such as sodium and magnesium to release hydrogen gas.



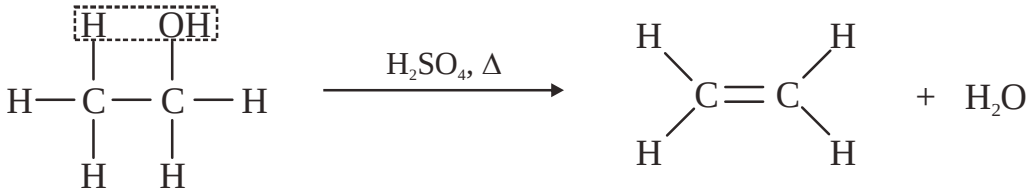
Figure–2: Reaction of alcohol with sodium metal

$$C_2H_5OH + Na \longrightarrow \underset{\textit{Sodium ethoxide}}{C_2H_5ONa} + H_2\uparrow$$

Similarly, methanol reacts with sodium to give sodium methoxide. What will happen when ethanol reacts with magnesium?

16.1.3.1 Dehydration of alcohols

When ethanol is heated at 443 K with excess of concentrated sulphuric acid the –OH group combines with the hydrogen on the nearest carbon atom to give water. Thus, the dehydration of alcohol leads to alkene formation.



16.1.3.2 Oxidation of alcohols

Alcohols are easily oxidized in the presence of strong oxidizing agents to form carboxylic acids.

$$C_2H_5OH \xrightarrow{\text{Oxidizing agent}} \underset{\textit{Ethanoic acid}}{CH_3COOH} + H_2O$$

$$CH_3OH \xrightarrow{\text{Oxidizing agent}} \underset{\textit{Methanoic acid}}{HCOOH} + H_2O$$

16.1.4 Uses of alcohols

1. Since many substances are soluble in alcohol therefore it is used as a solvent in paints, varnish etc.
2. The boiling point of methanol (methyl alcohol) is 65°C and it is highly inflammable therefore it is mixed with petrol and used as a fuel.
3. Ethanol is less harmful as compared to methanol therefore in most countries ethanol is mixed with petrol. Ethanol is also used as fuel in spirit lamps.
4. The freezing and melting point of alcohol is –114°C so it is used in low temperature thermometers. Alcohol is a colourless liquid that expands easily and uniformly. A red dye is added to it so that temperature can be read easily.
5. Glycol and glycerols are also types of alcohols which are used to make soaps, medicines and also used as antifreeze in car radiators.
6. Ethanol is used as a sterilizing agent in the medical field.

Harmful effects of alcohol

Consumption of small quantities of dilute ethanol causes intoxication. Long-term consumption of high quantities of alcohol can affect many vital organs such as the liver. Alcohol also slows down the metabolic processes in our body. Intake of very small quantities of methanol can cause death. It reacts rapidly with components of liver cells and poisons the entire body.

Ethanol is an important industrial solvent. It is manufactured on a large scale for industrial purposes. To prevent the misuse of ethanol, it is made unfit for drinking by adding poisonous substances like methanol (10%-15%) to it. Dyes are also added to colour the alcohol blue so that it can easily be identified that methanol has been added. This mixture is known as methylated spirit.

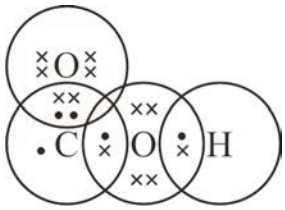
Questions

1. What are molasses?
2. What will happen when:
   (a) Ethanol is heated in the presence of concentrated sulphuric acid.
   (b) Yeast is added to molasses

16.2 Alkanoic Acids

You have read about acids in the chapter on acids, bases and salts. Carbonic acids are scattered all around us in nature, for example, acetic acid, citric acid, lactic acid etc. All these acids have the functional group –COOH which is known as the carboxylic group. Let us see the electron dot structure of –COOH.

When the carboxylic group is joined with the –H group then we get HCOOH which is known as formic acid. If instead of –H , we place –R (alkyl group) then we get RCOOH. In IUPAC nomenclature, these are called alkanoic acids, that is, the –e of alkane is replaced by –oic acid. Table-4 has some examples; use them to replace the question marks with the correct acid names.

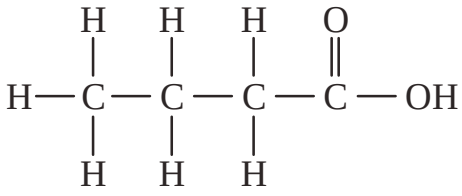


Structure of –COOH

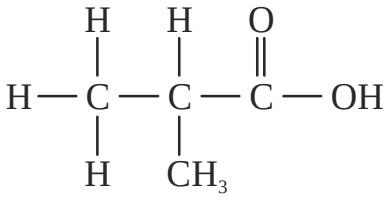
Table-4

Parent hydrocarbon	IUPAC name	Formula
Methan***e***	Methan***oic acid***	HCOOH
Ethane	?	CH_3COOH
Propane	?	?
?	Butanoic acid	?

You have read about isomerism in alcohol. Is it possible for alkanoic acids to have structural isomers? Will their names be different? Let us understand through some examples-



(A)



(B)

Just as we did for alcohols, in alkanoic acids also we must choose the longest, continuous carbon chain having the –COOH functional group. While numbering the carbon chain, the carbon atom to which –COOH group is attached is always given the number 1. In this way, the alkanoic acid shown by structure 'A' will be called butanoic acid and 'B' will be called 2–methylpropanoic acid.

When we look at structures 'A' and 'B' we find that both have the molecular formula $C_4H_8O_2$ but they are structurally different. Therefore, 'A' and 'B' are isomers of each–other. Look carefully at the two structures and tell whether 'A' and 'B' are chain isomers or structural isomers?

Questions

1. Write the structures of all possible isomers of $C_6H_{12}O_2$.
2. What is the IUPAC name of formic acid?

16.2.2 Industrial preparation of ethanoic acid (Quick vinegar method)

The industrial preparation of vinegar involves the fermentation of alcohol. In this method, a tank with holes at the bottom, to allow air–circulation, is used (figure-3). The tank is filled with wood shavings and mixed with ammonium sulphate. Ammonium sulphate is useful in increasing

bacteria population. Then, 10% ethanol is added slowly to the tank from above. Ethanoic acid is produced in the lowest portion of the tank by fermentation process. The ethanoic acid produced as a result is known as vinegar and the concentration of ethanoic acid in vinegar is 3-4%.

Pure ethanoic acid which is free from water is known as glacial acetic acid. Its boiling point is 118°C.

To test whether ethanoic acid has been produced or not, sodium carbonate is added to the product obtained. If acid has been formed then carbon dioxide gas would be released with rapid bubbling.

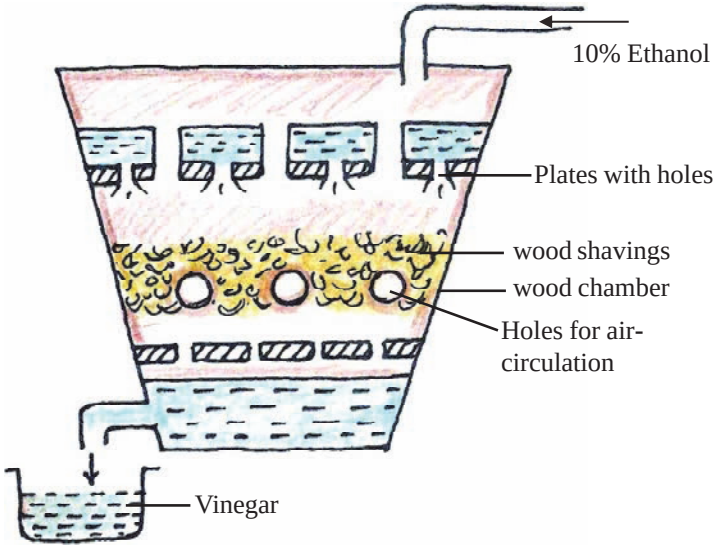


Figure-3: Quick vinegar method for preparing acetic acid

16.2.3 Properties of alkanoic acids

Ethanoic acid is acidic in nature and most acids are soluble in water. However, high molecular weight alkanoic acids (fatty acids), for example, citric acid are insoluble in water. Because of their acidic nature, they turn blue litmus red. Let us learn more about the chemical properties of some alkanoic acids:-

1. Salt formation: Since alkanoic acids are acidic in nature therefore they react with bases to give salt and water. For example, ethanoic acid (acetic acid) reacts with caustic soda to give sodium acetate salt.

$$CH_3COOH + NaOH \longrightarrow \underset{\textit{sodium acetate}}{CH_3COONa} + H_2O$$

Similarly, methanoic acid reacts with calcium hydroxide to give calcium formate.

$$2HCOOH + Ca(OH)_2 \longrightarrow \underset{\textit{calcium formate}}{(HCOO)_2Ca} + 2H_2O$$

Name the product formed when acetic acid reacts with magnesium hydroxide. Write the balanced chemical equation of the reaction.

Acetic acid reacts with sodium carbonate and sodium hydrogencarbonate to form salts with evolution of carbon dioxide gas.

$$CH_3COOH + Na_2CO_3 \longrightarrow CH_3COONa + H_2O + CO_2\uparrow$$

$$CH_3COOH + NaHCO_3 \longrightarrow CH_3COONa + H_2O + CO_2\uparrow$$

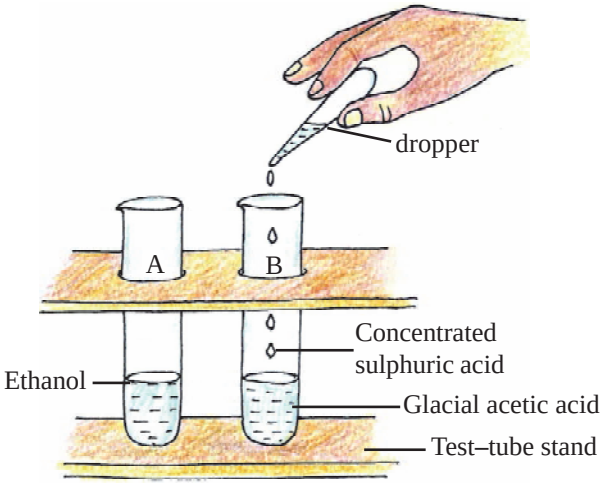
Balance the two equations given above. Can you think of some activity to test the presence of CO_2 gas?

2. Reaction with alcohol: When an alkanoic acid is heated with an alcohol in the presence of concentrated sulphuric acid, ester is formed as product. Esters have a distinct fruity smell and the reaction is known as esterification.

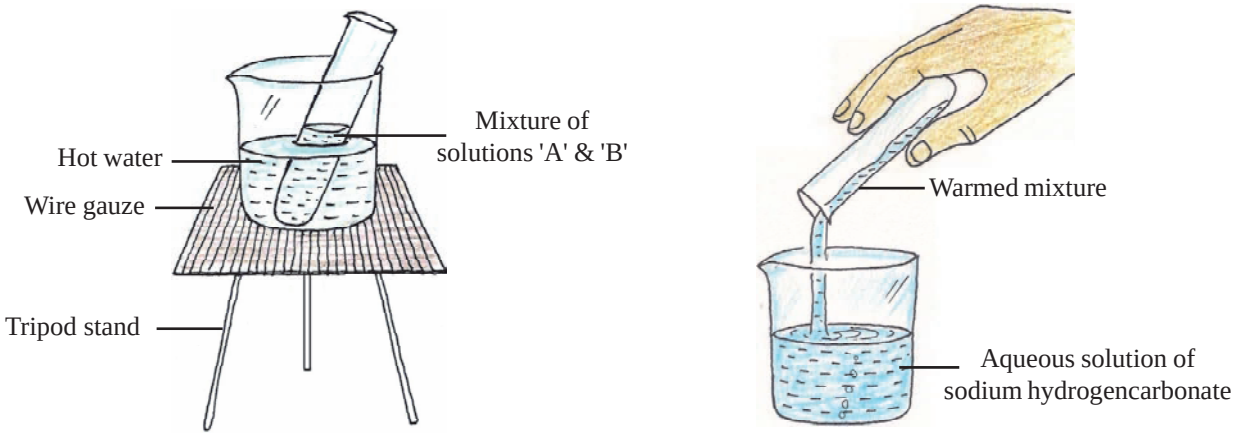
$$\underset{\textit{Ethanoic acid}}{CH_3COOH} + \underset{\textit{ethanol}}{CH_3CH_2OH} \xrightarrow{\text{Conc. sulphuric acid}} \underset{\textit{ester}}{CH_3COOCH_2CH_3} + H_2O$$

Activity–3

- Take 150 mL water in a beaker and warm it to 60°C.
- Take two test tubes and label them A and B respectively.
- Take 5 mL ethanol in test tube A.
- Take 5 mL glacial acetic acid in test tube B and add 4-5 drops concentrated sulphuric acid to it.
- Now add the solution in test tube A to the solution in test tube B.



- Now place the test tube in the warm water so that the reaction mixture also becomes warm. Keep shaking the test tube from time to time.
- Pour the reaction mixture in a solution of sodium hydrogencarbonate in water. Doing this will cause the unreacted ethanoic acid to react with $NaHCO_3$. Did you see any bubbles? (Figure-4).



Figure–4: Esterification

- Observe the reaction mixture and note if there is any distinct smell (sweet and fruity). The smell could be due to which substance?

3. Reaction with metals: Similar to other acids, alkanoic acids also reacts with metals such as sodium to release hydrogen gas.

$$CH_3COOH + Na \longrightarrow CH_3COONa + H_2\uparrow$$

Balance the equation given above. How will you test that the gas produced is hydrogen gas?

16.2.4 Uses of alkanoic acids

1. A dilute solution of acetic acid is known as vinegar and it is used as a preservative in pickles, sauces etc.
2. Alkanoic acids are used to make esters which are used in making perfumes, polymer etc.
3. Citric acid, tartaric acid and lactic acid are hydroxyl acids (that is, –OH functional group is also present along with –COOH group in these compounds). These acids are used as food preservatives.

Questions

1. What will happen when ethanol reacts with ethanoic acid? Write the balanced chemical equation of the reaction.
2. How can we make magnesium formate from formic acid? Show the reaction using a balanced chemical equation.

16.3 Polymers

In previous classes we saw that carbon compounds have chains of carbon atoms. Carbons atoms bond with each other to give compounds having very long chains, from two to thousand carbon atoms. Lower molecular weight carbonic molecules link to each other to give high molecular weight carbonic molecules and this process is known as polymerisation. The lower atomic weight mplecules are known as monomers and the high molecular weight molecule formed is known as polymer.In this way, each polymer has many monomers and their number is denoted by n. Proteins, carbohydrates, terylene, polythene etc. are some examples of polymers.

16.3.1 Natural and synthetic polymers

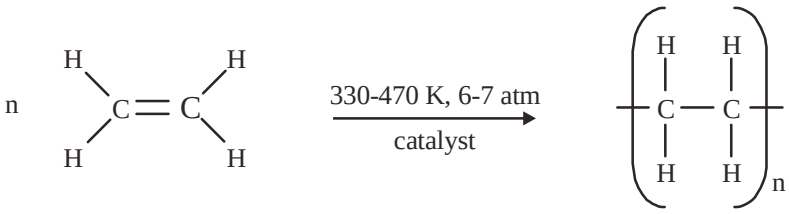
We obtain certain polymers such as wool and silk from animals and others such as starch and cellulose from plants. These polymers are known as natural polymers.

On the other hand, polymers made by human beings are called synthetic polymers. Polymers have made an immense difference in our daily lives and we use them in a variety of ways. Let us learn about some polymers.

16.3.2 Polythene

Polythene is a long chain polymer of the monomer ethane. There are two main methods of polymer production and the nature and properties of the polymers obtained by these methods are different.

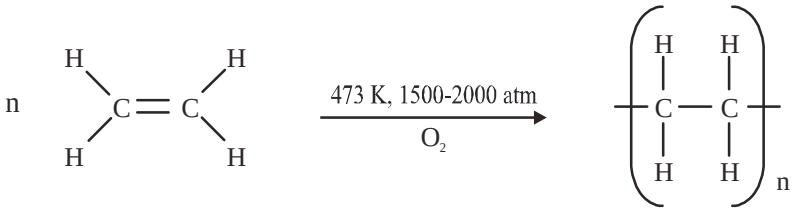
16.3.2.1 Low density polythene (LDPE): When a large number of ethene (ethylene) molecules are heated to 330-470 K under 6-7 atm pressure in the presence of a catalyst, we get a polythene which has a high melting point and low density.



Ethene *low density polythene*

This polymer is semi-transparent and hard and is therefore, used for making pipes and buckets. It is inert towards most chemicals.

16.3.2.2 High density polythene (HDPE): When a large number of ethene (ethylene) molecules are heated at 473 K in the presence of minute quantities of oxygen under 1500-2000 atmospheric pressure, we get a transparent polythene which is called high density polythene or HDPE.



Ethene *high density polythene*

Since this type of polymer has high density and low melting point, it is used for making bottles and sheets.

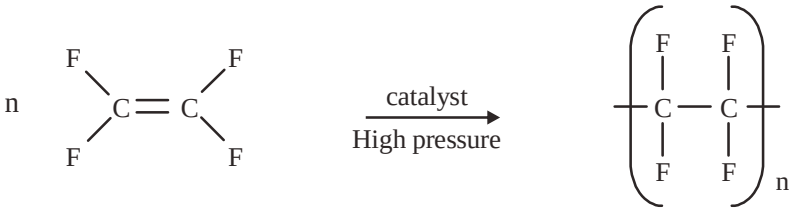
16.3.3 Teflon

Have you heard of cooking utensils in which the food does not stick when we cook it? What is the substance that is used to coat these utensils? This substance is actually the polymer, Teflon which is also used for making insulated instruments. Teflon is also used in making pipes or tubes that often come in contact with corrosive substances such as concentrated acids, concentrated alkalis, aqua regia. These days Teflon is also used in making packing gaskets (figure-4).



Figure–5: Packing gaskets

Teflon is made from tetrafluoroethylene at high pressures in the presence of a catalyst.



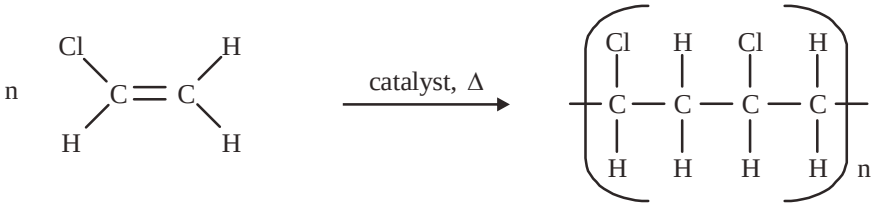
tetrafluoroethylene *Teflon*

Note that Teflon is inert to all chemicals but can melt at temperatures above 330°C. Therefore, while using Teflon coated utensils in cooking we should not cook at high flames.

16.3.4 Polyvinyl chloride

Have you ever wondered what is common to the material used to make raincoats, table covers, mudguards of cycles and motorcycles, toys, gloves and electrical–wire coatings?

You will find that all these materials are electrical insulators and made of polyvinyl chloride (PVC). To produce PVC, vinyl chloride is heated in the presence of a catalyst. PVC is also inert to heat and chemicals.



Vinyl chloride *polyvinyl chloride*

16.3.5 Harmful effects of polythene

Till very recently, most of the materials and objects that we used in our day to day lives such as furniture, paper bags, jute fabrics etc. were made from plant-products. These were easily decomposed although cutting trees in excess to obtain products caused environmental imbalance.

We started using polythene, which is a petroleum product, thinking that this would reduce the cutting of trees. We now know that most chemicals do not affect polythene and so it cannot be decomposed easily. Therefore, it is rapidly becoming a cause of environmental pollution.

Because polythene does not decompose, it affects not only the lives of humans but the entire ecological system. If we try to destroy it by burning, gases such as CO_2 and CO are produced. It is clear from figure-6 that regular use of polythene leads to pollution of environmental factors such as water, air and soil. When animals eat polythene they fall sick and can even die because they are unable to digest it. So we see that polythene affects not just non–living aspects but living beings as well. We humans have also been harmed by use of polythene.



Figure–6: Environmental effects of polythene

In this situation, it is of primary importance that we eliminate or minimize the use of polythene. Several positive steps have been taken in this direction. Research shows that some microorganisms are capable of decomposing polythene but more research is still needed in this area. It is also proposed that polythene can be melted to build roads. Many other initiatives have been undertaken to recycle polythene.

Keywords

Functional group, alcohol, methanol, ethanol, fermentation, enzyme, absolute alcohol, rectified spirit, methylated spirit, carboxylic acid, ethanoic acid, glacial acetic acid, monomer, polymer, polymerization, antifreeze, ester, esterification, PVC, low density polyethylene, high density polyethylene, Teflon



What we have learnt

- ROH and RCOOH are used to denote alcohols and acids respectively.
- An atom or group of atoms which when joined to an alkyl group determines the characteristic chemical properties of that carbonic compound, is known as a functional group.
- Alkyl groups are depicted by –R.
- The functional group in alcohols is –OH and that in alkanoic acids is –COOH.
- Fermentation of molasses is used for industrial preparation of alcohol.
- Pure alcohol which does not have any water is known as absolute alcohol.
- Rectified spirit is ethanol having 5% water.

- To prevent the misuse of ethanol, it is made unfit for drinking by adding poisonous substances like methanol (10%-15%) to it. This mixture is known as methylated spirit.
- Fermentation of ethanol is used to prepare ethanoic acid.
- 3-4% solution of ethanoic acid is known as vinegar.
- Pure ethanoic acid is known as glacial acetic acid.
- Low molecular weight monomers combine together to give high molecular weight polymers. This process is known as polymerization.
- Polymers like silk and wool which are obtained from animals and those like starch and cellulose which are obtained from plants are called natural polymers. Those polymers that are made by humans are called synthetic polymers.
- The monomer of polythene is ethene. The polymer of vinly chloride is called polyvinylchloride. The monomer of teflon is tetrafluroethene.

Exercise

1. Choose the correct option

   (i) When ethanol has 5% water added to it, it is called-

   (a) Rectified spirit (b) Methylated spirit

   (c) Methanol (d) Absolute alcohol

   (ii) Packing gasket is made of

   (a) Low density polythene (b) High density polythene

   (c) teflon (d) Polyvinyl chloride

   (iii) If sodium carbonate is added to ethanoic acid, a gas is evolved with rapid bubbling. This gas is-

   (a) CO (b) CO_2

   (c) O_2 (d) Water vapour

   (iv) The IUPAC name of acetic acid is-

   (a) Ethanoic acid (b) ethanol

   (c) Methanol (d) Methanoic acid

2. What happens when-

   (i) Ethanoic acid reacts with sodium

   (ii) Ethanol reacts with sodium

3. Complete the following chemical eqautions:

   (i) $C_2H_5COOH + ? \longrightarrow (C_2H_5COO)_2Mg + H_2$

   (ii) $CH_3OH + CH_3COOH \longrightarrow ? + H_2O$

   (iii) $CH_3COOH + Na_2CO_3 \longrightarrow ? + ? + CO_2$

4. Why is ethanol used in thermometers?

5. How is ethene obtained from ethanol?

6. Write the structures of possible isomers of $C_5H_{10}O_2$.

7. How is ethanol manufactured industrially?

8. What is vinegar? How is it manufactured?

9. What will be the condensed structure of PVC? Write any four uses of PVC.

10. Nadira was studying the chemical properties of ethanol and ethanoic acid. To do this, she took ethanoic acid and concentrated sulphuric acid in a testtube and heated the mixture. While she was carrying out this activity, Meena came and added ethanol to the mixture. When this mixture was added to water, there was a unique odor. Can you tell which chemical reaction took place and write the balanced equation for the reaction.

11. Abraham had two solutions - ethanol and ethanoic acid. He added sodium to both the solutions and saw that a gas was produced. When a burning matchstick was placed near the gas, a pop sound was heard. Show the reactions through chemical equations.

12. Jaspreet has ethanol and ethanoic acids in two separate beakers but she forgot to label them. Help Jaspreet identify which beaker has ethanol and which has ethanoic acid.

13. What is a polymer? Explain the polymerization of ethene.

14. What is the difference between low density polythene and high density polythene? Write their used based on their properties.

15. When a carbon compound, A having molecular weight 46u was oxidized, a carbonic acid B was obtained. Compound A is used a sterilizing agnet. Identify A and B and write the balanced chemical equations.